

Impact of Initialization on Clustering Performance

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04.09.2026

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1 Introduction

The objective of this work is to study the impact of centroid initialization on the performance and stability of K-Means clustering. K-Means is an iterative clustering algorithm whose final solution depends on the initial positions of its centroids. Consequently, different initializations can lead to different local optima, resulting in different clustering quality and convergence behaviour.

Two initialization strategies were compared:

1. Standard K-Means with randomly selected initial centroids;
2. K-Means++ initialization, which selects the first centroid randomly and subsequently selects additional centroids with probability proportional to their squared distance from the closest already selected centroid.

The comparison was performed through repeated trials. The primary performance measure was inertia, defined as the sum of squared distances between each observation and the centroid of its assigned cluster. Variability of the resulting inertia values was also examined. In addition, clustering stability was investigated by comparing cluster assignments between different trials.

Three experimental settings were considered: a synthetic dataset with relatively clear clusters, a synthetic dataset with overlapping clusters, and the MNIST handwritten-digit dataset as a real-world high-dimensional dataset.

2 Experimental Setup

2.1 Algorithms

Both clustering algorithms were implemented from scratch rather than using the corresponding implementation from a machine-learning library.

Standard K-Means

The standard K-Means implementation begins by randomly selecting k observations from the dataset as initial centroids. Each iteration consists of three main operations:

1. Assign every observation to its nearest centroid.
2. Recalculate each centroid as the mean of the observations assigned to it.
3. Check whether the centroid movement is below the specified convergence tolerance.

The implementation uses Euclidean distance for assigning observations to clusters. Empty clusters are handled by reinitializing their centroid using a randomly selected observation. The algorithm has a maximum of 100 iterations and uses a convergence tolerance of 10^{-4} .

K-Means++

The second implementation uses K-Means++ initialization while keeping the subsequent K-Means optimization procedure essentially the same. The first centroid is selected randomly. For each subsequent centroid, the squared distance of every observation to its nearest existing centroid is calculated. The next centroid is then sampled with probability proportional to this squared distance. This encourages the initial centroids to be distributed across different regions of the dataset.

After initialization, the algorithm follows the same assignment and centroid-update procedure as standard K-Means. This design isolates initialization as the principal difference between the two methods.

3 Datasets

3.1 Synthetic Dataset with Clear Clusters

The first dataset was generated using make-blobs. It contains:

- 300 observations
- 2 features
- 5 clusters
- cluster standard deviation of 0.8

The random seed is fixed at 42 for dataset generation. A synthetic dataset was selected because its structure is controlled and relatively easy to visualize. The low-dimensional representation also makes it possible to display the observations and centroids directly in two-dimensional plots. This

dataset represents a favourable situation for K-Means: clusters have a relatively clear structure and can be reasonably represented by their centroids.

3.2 Synthetic Dataset with Overlapping Clusters

The second synthetic experiment investigates the behavior of initialization when clusters are less clearly separated. Again, make-blobs was used, but the standard deviation was increased to 2.5. The dataset contains:

- 300 observations
- 2 features
- 4 clusters
- cluster standard deviation of 2.5

Increasing the cluster standard deviation produces greater overlap between clusters. This makes the clustering problem more difficult and allows the experiment to investigate whether the advantage of K-Means++ remains when the underlying cluster structure is less distinct.

3.3 MNIST

The third experiment uses MNIST, a real-world dataset of handwritten digits. A subset of 1,000 observations is selected. The original pixel values are normalized by dividing them by 255. PCA is subsequently applied, reducing the representation to 50 principal components. Thus, K-Means is performed on a 50-dimensional representation rather than directly on the original 784-dimensional pixel space. The number of clusters is set to 10, corresponding to the ten digit classes.

PCA was used both to reduce dimensionality and to make the high-dimensional experiment computationally more manageable. The code also reports the cumulative explained variance of the 50 principal components. It is important to note that this experiment should be described as MNIST after PCA dimensionality reduction, rather than clustering directly in the original 784-dimensional space.

4 Repeated-Trial Experimental Design

The central part of the experiment is the use of multiple independent initializations. For every trial, standard K-Means and K-Means++ are executed using the same trial number as their random seed. The synthetic experiments use 50 trials, while the MNIST experiment uses 30 trials.

For every trial, the following quantities are stored:

- final inertia
- number of iterations until convergence
- final cluster labels

This produces a distribution of outcomes rather than relying on a single K-Means run. This is particularly important because the purpose of the experiment is to investigate the influence of initialization. A single run would provide very little evidence about initialization variability.

5 Evaluation Measures

5.1 Inertia

Lower inertia indicates that observations are, on average, closer to their assigned cluster centroids. The implementation explicitly calculates the sum of squared distances from observations to their corresponding final centroids.

For each algorithm, the experiment calculates:

- mean inertia
- standard deviation of inertia
- median inertia
- minimum inertia
- maximum inertia
- coefficient of variation

The percentage reduction in mean inertia and standard deviation between standard K-Means and K-Means++ is also reported.

5.2 Variability

Variability is evaluated primarily through the distribution of final inertia values across repeated trials. A small standard deviation indicates that the algorithm tends to converge to solutions with similar inertia despite different initializations. Conversely, a large standard deviation indicates greater sensitivity to initialization.

5.3 Clustering Stability

The code additionally attempts to evaluate clustering stability by comparing the labels generated by different trials. For every pair of trials, the proportion of observations receiving the same numerical cluster label is calculated. The resulting pairwise agreement values are visualized as distributions.

However, this measure has an important limitation. Cluster labels are arbitrary identifiers. Two runs can produce exactly the same partition while assigning different numbers to the clusters. Therefore, raw label agreement is not a fully valid permutation-invariant clustering stability metric. The resulting stability analysis should consequently be interpreted cautiously. The experiment nevertheless provides an indication of how consistently the numerical cluster assignments are reproduced across trials.

6 Visualisations

Several visualisations are generated for each experiment.

6.1 Inertia Distribution

Histograms show the distribution of final inertia values for K-Means and K-Means++. This visualization is particularly relevant because it shows both the typical performance of each algorithm and the spread of possible outcomes. A narrow distribution concentrated around low inertia values

indicates relatively stable performance, while a wider distribution indicates stronger dependence on initialization.

6.2 Convergence Iterations

A second histogram shows the distribution of the number of iterations required for convergence. This provides an additional perspective on the algorithms. K-Means++ may require a different number of optimization iterations because its starting centroids are generally more deliberately distributed. The number of iterations is not itself a direct measure of clustering quality, but it provides useful information about convergence behavior.

6.3 Inertia Box Plot

A box plot summarizes the inertia distributions for both initialization methods. It makes it easier to compare central tendency, spread, possible outliers, and relative variability.

6.4 Inertia Across Trials

The code also plots the inertia obtained in every individual trial. This visualization makes the effect of different random initializations directly visible. If standard K-Means exhibits large fluctuations between trials while K-Means++ produces more tightly grouped values, this provides visual evidence that K-Means++ is less sensitive to initialization.

6.5 Stability Distribution

Pairwise label agreement values are displayed as histograms. The purpose is to compare how consistently each algorithm reproduces cluster assignments across different trials. As discussed above, the interpretation of this visualization is limited by the fact that the metric does not account for permutations of cluster labels.

6.6 Summary Statistics

A final panel displays numerical summary statistics for the two algorithms, including mean inertia, standard deviation, minimum inertia, and mean number of iterations. The code also reports the percentage reduction in mean inertia and standard deviation obtained with K-Means++.

7 Analysis of Expected Results

7.1 Clear Synthetic Clusters

The clear-cluster synthetic dataset provides the simplest environment for examining initialization. Because the clusters have relatively distinct structure, both algorithms should generally be capable of finding good solutions. Consequently, the difference between K-Means and K-Means++ may not be dramatic in every trial.

Nevertheless, random K-Means can occasionally select several initial centroids from the same region. This can cause some clusters to be poorly represented initially and may result in convergence to a less favourable local optimum. K-Means++ addresses this problem by encouraging the initial centroids to be spread across the dataset.

Therefore, the main positive result expected from this experiment is not necessarily that K-Means++ always obtains a dramatically lower inertia, but that it should generally produce more reliable initial configurations and fewer poor solutions. The repeated-trial design allows this behaviour to be observed through the distributions rather than through a single example.

7.2 Analysis of Overlapping Clusters

The overlapping-cluster experiment is more challenging. When cluster distributions overlap substantially, the distinction between clusters becomes less clear. Consequently, even a good initialization cannot completely eliminate the ambiguity inherent in the data.

This provides an important negative aspect of K-Means++: better initialization cannot compensate for unsuitable or ambiguous cluster structure. In this setting, both algorithms may produce greater variation in inertia and cluster assignments. K-Means++ can still provide better starting points, but the advantage may be smaller because the underlying clustering problem itself is difficult.

This experiment therefore demonstrates an important distinction: initialization affects the optimization process, but it does not change the information contained in the dataset. If clusters overlap strongly, multiple partitions can be plausible, and K-Means++ cannot guarantee a uniquely correct clustering.

7.3 Analysis of MNIST

MNIST provides a substantially more complex setting than the two-dimensional synthetic datasets. The observations are represented using 50 PCA components, and ten clusters are requested. Unlike the synthetic data, handwritten digits can have substantial intra-class variation and visual similarity between different digits. Consequently, the K-Means objective may have many different local optima.

The repeated trials are therefore particularly useful here. Comparing the distributions of inertia between K-Means and K-Means++ allows the influence of initialization to be investigated under more realistic and high-dimensional conditions.

A positive result would be a lower and less variable inertia distribution for K-Means++. A negative result would be a relatively small difference between the algorithms, demonstrating that initialization is only one factor affecting clustering performance on complex real-world data. The PCA preprocessing also introduces a limitation: the experiment does not measure the behaviour of K-Means directly in the original 784-dimensional representation.

8 Positive Aspects of the Experiment

Several aspects of the implementation strengthen the experiment.

8.1 Direct comparison of initialization methods

The two algorithms use essentially the same K-Means optimization procedure, with initialization being the principal difference. This makes the comparison relevant to the stated research question.

8.2 Repeated trials

Using 50 trials for the synthetic datasets and 30 trials for MNIST is substantially more informative than relying on a single random initialization.

8.3 Multiple datasets

The experiment does not rely exclusively on an artificial dataset. It combines clearly separated synthetic clusters, overlapping synthetic clusters, and real-world high-dimensional MNIST data (after PCA). This makes it possible to examine initialization under different levels of difficulty.

8.4 Multiple evaluation perspectives

The experiment does not rely exclusively on mean inertia. It examines distributions, standard deviation, coefficient of variation, convergence iterations, and trial-to-trial behaviour.

8.5 Explicit implementation

Implementing K-Means and K-Means++ directly makes the difference between their initialization procedures explicit and easier to study.

9 Negative Aspects and Limitations

The experiment also has several limitations.

9.1 Stability metric

The most important limitation is the use of raw label agreement for stability. Because cluster IDs can be permuted between runs, numerical equality of labels does not necessarily mean that two clusterings are similar. Therefore, this measure can underestimate actual clustering stability. A permutation-invariant measure such as Adjusted Rand Index (ARI) or Normalized Mutual Information (NMI) would provide a more appropriate stability analysis. This limitation should be explicitly acknowledged when interpreting the stability plots.

9.2 MNIST dimensionality reduction

Although MNIST is intrinsically a 784-dimensional dataset, the actual clustering is performed after PCA reduction to 50 dimensions. Thus, the experiment investigates high-dimensional data after dimensionality reduction rather than the full original feature space.

9.3 Inertia does not measure semantic correctness

Inertia only measures compactness with respect to the K-Means objective. A lower inertia does not necessarily mean that the resulting clusters correspond better to the actual digit classes. The

MNIST labels are loaded by the program, but they are not used to calculate an external clustering-quality metric such as ARI against the ground-truth labels. Consequently, the experiment evaluates optimization performance rather than semantic clustering accuracy.

9.4 Fixed number of clusters

The number of clusters is manually specified: $k = 5$ for the clear synthetic dataset, $k = 4$ for the overlapping dataset, and $k = 10$ for MNIST. The experiment therefore does not investigate the problem of selecting the appropriate value of k .

9.5 Computational efficiency

The custom K-Means and K-Means++ implementations calculate distances using explicit NumPy operations and, in the K-Means++ initialization, repeatedly calculate distances for observations. This is suitable for an educational experiment but is less optimized than specialized production implementations.

10 Overall Interpretation

The experiment is designed to demonstrate an important property of K-Means: the final clustering solution can depend substantially on initialization.

Standard K-Means selects initial centroids randomly. As a result, repeated runs on the same dataset can converge to different local optima. This manifests itself as variation in final inertia and potentially in the resulting cluster assignments.

K-Means++ uses a more informed initialization strategy. By favouring points that are far from already selected centroids, it attempts to distribute the initial centroids across different regions of the dataset.

The repeated experiments provide a way to examine whether this produces:

- lower average inertia
- lower variability in inertia
- fewer poor solutions
- more consistent convergence behaviour

The code additionally calculates the probability that a run obtains a solution within 10 percent of the best K-Means++ solution. This provides another way to compare the likelihood of obtaining a good solution across the two initialization strategies.

Overall, K-Means++ should be interpreted not as a different clustering algorithm after initialization, but primarily as a better initialization strategy for the same underlying K-Means optimization problem.

11 Conclusion

The implemented experiment investigates the effect of initialization on K-Means clustering using repeated trials across datasets of increasing difficulty.

The first synthetic dataset provides a controlled environment with relatively clear clusters. The second introduces overlapping clusters to investigate the effect of more ambiguous cluster structure. Finally, MNIST provides a real-world high-dimensional dataset, with PCA reducing the representation to 50 dimensions before clustering.

Both standard K-Means and K-Means++ are implemented explicitly, and their performance is evaluated over multiple random trials. Inertia distributions, variability, convergence iterations, and trial-to-trial clustering behaviour are visualized and summarized statistically.

The principal expected advantage of K-Means++ is improved robustness to initialization. Instead of relying entirely on randomly selected starting centroids, K-Means++ deliberately spreads the initial centroids across the dataset. This should reduce the probability of poor initial configurations and, consequently, poor local optima.

At the same time, the experiments demonstrate that initialization is not the only determinant of clustering quality. Overlapping clusters and complex high-dimensional data can remain difficult even with improved initialization.

The implementation therefore successfully demonstrates the central idea of the experiment: initialization can have a substantial effect on K-Means performance, and K-Means++ provides a principled strategy for reducing this dependence. However, the stability analysis should be interpreted with caution because the implemented label-agreement measure does not account for arbitrary permutations of cluster labels.